

Introduction to Data Science Project -2

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Topic Proposal

We will use the NHANES “What We Eat in America” dataset (used in previous [project](#)) to study how demographic and lifestyle variables shape dietary patterns in the United States. We will investigate nutrient intake behaviour and build predictive regression models that help explain what drives healthier versus less healthy diets across the U.S. population.

We will estimate three models, each aligned with a specific SMART question:

Model 1: Linear Regression – Determinants of Sodium Intake

- SMART Question: Which demographic and socioeconomic variables predict sodium intake, and how strongly?
- Focus variables: Income, length of time in the U.S., age, gender, ethnicity.
- Goal: Create a predictive model for sodium intake, test interactions, evaluate transformations for normality, use survey weights.

Model 2: Linear Regression – Predicting Sugar Intake by Meal Context

- SMART Question: Do demographic and lifestyle characteristics predict sugar intake differently for total daily consumption vs only snack-time consumption?
- Goal: Model sugar intake using two specifications (daily total vs snack-only subset), compare coefficients, consider outlier control/transformations, test model fit.

Model 3: Logistic Regression – Predicting a “Healthy Diet” Classification

- SMART Question: Can demographic and nutritional variables predict whether an individual meets recommended daily nutrition benchmarks?
- Outcome variable: Healthy = meets daily intake thresholds for calories, sodium, protein, sugar, carbs, fiber, and fat.
- Goal: Estimate a logit model, evaluate predictive accuracy (ROC, confusion matrix), explore key predictors.

Dataset Details:

- National Center for Health Statistics (NCHS)
 - NHANES (National Health and Nutrition Examination Survey) 2021-2023
 - WWEIA (What We Eat in America)
 - Collected as part of NHANES to monitor the food and nutrient intakes of the U.S. population.
- Files
 - Files include DEMO (demographics), DR1IFF (Day 1 foods), DR1TOT (Day 1 totals), and many others.
 - [Link to data](#)